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$$\begin{aligned}f(x) &= \text{Bgtan}\left(\frac{1}{x}\right) \\f'(x) &= \frac{1}{1 + \left(\frac{1}{x}\right)^2} \cdot \frac{0 \cdot 1 - 1 \cdot 1}{x^2} \\&= \frac{1}{1 + \frac{1}{x^2}} \cdot \left(-\frac{1}{x^2}\right) \\&= \frac{1}{\frac{x^2}{x^2} + \frac{1}{x^2}} \cdot \left(-\frac{1}{x^2}\right) \\&= \frac{1}{\frac{x^2 + 1}{x^2}} \cdot \left(-\frac{1}{x^2}\right) \\&= \frac{x^2}{x^2 + 1} \cdot \left(-\frac{1}{x^2}\right) \\&= -\frac{x^2 \cdot 1}{(x^2 + 1) \cdot x^2} \\&= -\frac{1}{x^2 + 1} \\f''(x) &= -\frac{0 \cdot (x^2 + 1) - (1 \cdot 2x)}{(x^2 + 1)^2} \\&= \frac{2x}{(x^2 + 1)^2}\end{aligned}$$

References